What a dissipative quantum system remembers

Markus Baumann [1,2], Itamar Fink [2], Johannes Wittmann [2], Claudia Linnhoff-Popien [1,2], and Jonas Stein [1,2]

[1]Quantum Applications and Research Laboratory (QAR-Lab), Ludwig-Maximilians-Universität München, Munich, Germany
[2]Mobile and Distributed Systems Group, Ludwig-Maximilians-Universität München, Munich, Germany

Dissipation has emerged as a resource for quantum reservoir computing. Here we extend this perspective by asking how its form determines what a reservoir can remember. We show that changing how the system couples to its environment can reverse the effect of stronger loss, allowing it to prolong memory. We prove a law describing how subsequent inputs mix with the stored record, linking this accumulation to memory lifetime and input disturbance. The law also applies to classical reservoirs under its stated assumptions. Writing more gently as memory grows more than doubles the capacity to recall individual inputs in our simulations without measurement noise. Finite measurements impose a further constraint: stored information can become too faint to recover. Together, these results establish a quantitative basis for designing dissipation, input, and measurement jointly to preserve information in a form that remains useful during computation.

1 Introduction

Processing information over time requires a memory of what came before. A physical system can provide such a memory because each input changes its state and leaves traces that influence its later response. Reservoir computing uses these traces to perform a task: an input stream drives a dynamical system, and a weighted sum of quantities describing its state produces the output. Only these weights are trained; this mapping is called a linear readout [1, 2]. In quantum reservoir computing (QRC), the evolving system is quantum mechanical, and the readout uses quantities obtained by measuring it [3]. The challenge is to keep earlier inputs accessible while new information continues to arrive.

The environment shapes how these traces fade. A system can lose energy or information to its surroundings through processes collectively described as dissipation [4–14]. Previous work has shown that tuning the dissipation rate can improve memory, sometimes yielding the best performance at an intermediate loss rate [6, 8]. Input, interactions, and dissipation can also be optimized together [15, 16]. The rate specifies how strongly the environment acts, but the way it acts is a separate design choice. Engineering this interaction can prepare and protect selected quantum states [17–20]. For a system processing an ongoing input stream, this raises the possibility of controlling which traces of the past survive.

This distinction becomes concrete for a reservoir of interacting spins, each a two-level quantum system. When the spins decay through separate routes into the environment, they undergo independent decay. When they share a common route, they undergo collective decay. In certain joint states, the different emission paths interfere and cancel, preventing decay through that shared route. These protected states are called dark states [21–23]. Internal dynamics can still move the system into states that do decay. If this decay is sufficiently rapid, however, it suppresses those transitions, a phenomenon known as the dissipative quantum Zeno effect [24–27]. Stronger loss can therefore slow the escape of information from the protected states (Fig. 1c).

Information that survives can nevertheless become difficult to use. Each new input changes the same state that stores the past. A signal that initially records one input may therefore come to depend on combinations of several inputs, such as their product. This mixing is useful for tasks

Corresponding author
Markus Baumann markus.baumann@campus.lmu.de

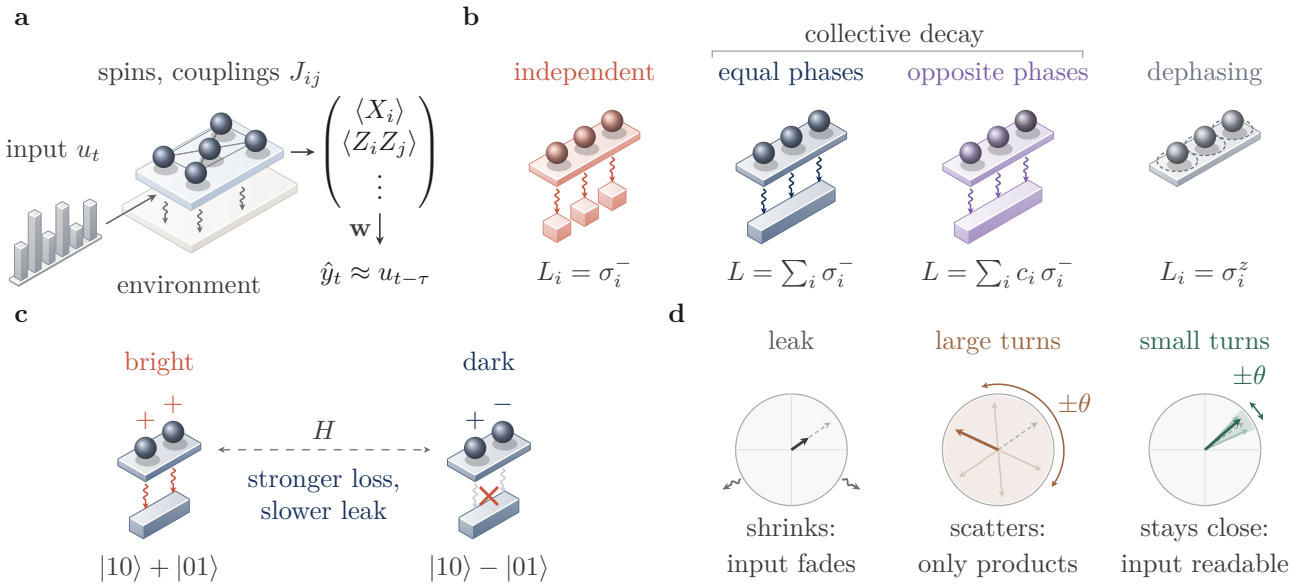


Figure 1: **Engineering how a quantum reservoir leaks. a** A stream of inputs drives interacting spins, and a trained linear readout recalls past inputs from the spins' expectation values. **b** Four forms of dissipation with the same total coupling. Each spin decays on its own (independent), all spins decay through one shared channel (collective decay with equal phases), the darker spins couple to that channel with the opposite sign (opposite phases), or the spins lose only their phase (dephasing). **c** Two spins share one channel and one excitation. If the excitation sits on both spins with the same sign, their decays add and the state decays fast (bright). With opposite signs they cancel, and the state cannot decay (dark). When the two spins differ, their own dynamics (H) drives them out of the dark state, but strong loss acts like a measurement repeated so often that they hardly leave it: the quantum Zeno effect. **d** Two ways to forget the record of one input, drawn as an arrow, dashed where it was written. It shrinks as it leaks away, or later inputs turn it at random. Large turns scatter it until only products of inputs remain, while small turns keep the input readable on its own.

that require relationships between inputs, but it reduces the capacity available for recalling individual inputs [28–30]. The lifetime of stored information and the time over which an individual input remains readable are therefore different quantities. Previous work has established memory limits in classical reservoirs [2, 31, 32], explored input strength [33], and investigated how noise and protected states affect quantum information processing [34–37]. The question is how protection, continued input, and measurement together determine the part of the past that remains recoverable.

Here we establish this connection in dissipative spin reservoirs (Fig. 1). We compare decay processes at the same total coupling strength to the environment, keeping the internal interactions fixed. Increasing independent decay shortens memory, whereas collective decay supports a regime in which stronger loss prolongs it. We further show that the way the spins participate in the shared decay process determines how readily an input can change the protected part of the state. The environment thus controls both how information survives and how new information can be written.

We then prove a law for the accumulation of input mixing. Under explicit mixing assumptions and for long memories, the law relates the number of distinct inputs combined in the stored record to the memory lifetime and the disturbance caused by each new input. It motivates a practical principle: reduce how strongly each input changes the system as memory grows. We call this gentle writing. In our spin reservoirs, it more than doubles the capacity to recall individual past inputs when measured quantities are known without statistical error. The improvement reallocates existing capacity from combinations of inputs towards individual inputs; the total capacity available to the readout remains unchanged.

Finally, we include the resources required to recover the stored information. Resolving a weak quantum signal requires repeated preparation and measurement to distinguish it from statistical fluctuations [9, 12, 38]. A long-lived record can therefore be difficult to retrieve if its signal is too faint. At a fixed measurement budget, weak

independent decay with gentle writing recovers the most memory in our comparison. The protected memories can persist just as long, but their weaker signals require more measurements. The available resources consequently change which environmental interaction and input strength are most useful.

Together, these results establish a quantitative connection between the physical survival of information and its usefulness during continued computation. They show how dissipation can be structured to preserve a memory that remains writable, how input strength should be matched to its lifetime, and how measurement resources constrain what can be recovered. The protection mechanism relies on quantum interference, while the mixing law also applies to classical reservoirs under the same assumptions. The resulting design principle extends beyond a particular decay process: the environment, the input, and the readout must be coordinated around the information that remains accessible. This makes recoverable memory a quantitative target for engineering physical systems that process information over time.

2 Model

The reservoir is the driven spin system of Ref. [6]. Five spins ($N = 5$) interact through random couplings J_{ij}, and in each step a random input u between 0 and 1 sets the strength of a transverse field for a time Δt:

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i, \tag{1}$$

with Pauli operators X_i, Z_i of spin i and a field strength h. To weaken the input at a fixed mean drive, we replace $h(1+u)$ by $h[\frac{3}{2} + \eta(u - \frac{1}{2})]$, where the amplitude η scales how far each input moves the field. The state ρ of the spins follows a Lindblad master equation [39, 40],

$$\dot{\rho} = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big), \tag{2}$$

where the Hamiltonian drives the internal dynamics and each jump operator L_k, scaled by one loss rate γ, is one way the environment changes the state, such as removing an excitation or erasing a phase.

A linear readout of 45 features, the single-spin and two-spin expectation values, is trained to estimate the input from τ steps back. Its recall C_τ, between 0 and 1, is the squared correlation between the estimate and the input on held-out data. The linear memory capacity $\sum_\tau C_\tau$ adds up the recall over all delays. One unit is one past input recalled perfectly, and the ceiling is 45, the number of features. The memory lifetime t_{mem} says how long the variance of a stored signal survives, and the single-input recall time t_{f} how long one input stays linearly readable on its own (Appendices A.3 and A.4). Sections 3.1 to 3.3 use exact expectation values, as if every measurement were repeated infinitely often, and Sec. 3.4 counts measurements.

We compare decay channels at equal total coupling $\mathrm{Tr}\, K = \sum_k \mathrm{Tr}\, L_k^\dagger L_k$ and hold the Hamiltonian fixed to isolate the effect of the dissipative structure. The generator of Eq. (2) has 4^N modes, patterns of change in the state that each decay at their own rate. Equal total coupling fixes their mean rate for every Hamiltonian, but not the slow modes, which keep a record longest and can become slower or faster (Appendix A.2). Table 1 lists the eight channels we studied.

Four of them carry the argument (Fig. 1b). With independent decay every spin loses its excitation on its own. With collective decay all spins lose it through one shared channel, either with equal phases or with opposite phases on spins 4 and 5. We call these the equal-phase and the opposite-phase channel. With dephasing every spin loses its phase. What tells them apart is whether the dissipation leaves only a single stationary state or a protected subspace that can carry a record. Operators that the dissipation alone leaves unchanged are protected, although the Hamiltonian can still turn them or couple them to decaying ones. Independent decay protects only its ground state. Neither collective channel can empty its ten dark states, so every operator on them is protected. Dephasing keeps the populations, the probabilities of the spin configurations, but erases the coherences between them. Every step then leaves the fully mixed state, with equal populations and no coherence, unchanged whatever the input: the dynamics is unital. The driven reservoir relaxes to this state, after which no input leaves a trace, so dephasing stores nothing.

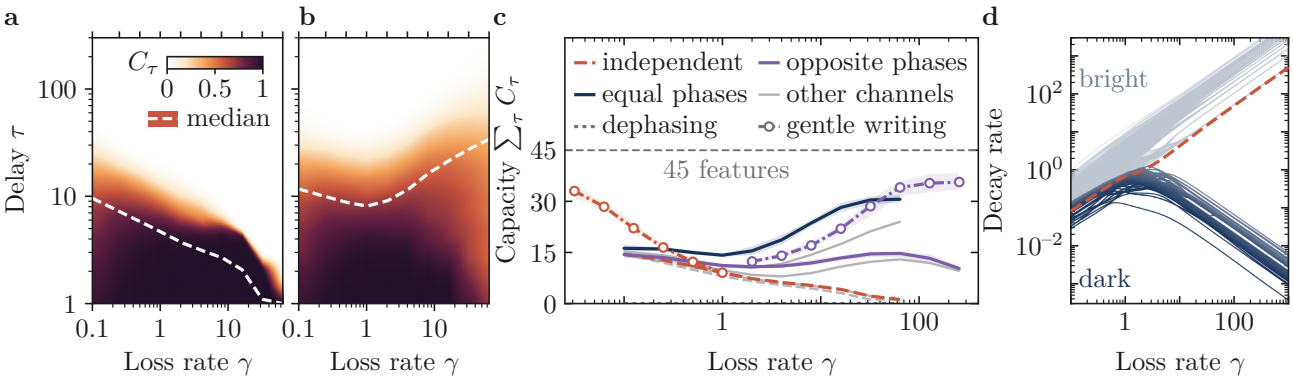


Figure 2: **The decay structure decides whether loss shortens or prolongs the memory. a,b** Recall of the input τ steps back (colour) against the loss rate, for independent decay (a) and the equal-phase channel (b). The dashed line marks the median delay, which splits the recall in half. **c** Linear memory capacity at the full input strength and with gentle writing, an input that weakens as the memory lifetime grows (Sec. 3.3). **d** All decay rates with the equal-phase channel: the dark modes slow down as the loss grows, while the slowest mode of independent decay (dashed) speeds up.

3 Results

3.1 Dissipative structure controls memory lifetime

Increasing the loss separates the channels into distinct memory regimes (Fig. 2). With independent decay, recall retreats to ever shorter delays and the linear memory capacity falls steadily (Fig. 2a,c). With the equal-phase channel the capacity first dips and then doubles, and at strong loss inputs fifty steps back are still recalled (Fig. 2b,c). The other channels split the same way (Fig. 2c, grey): exchange and gain/loss follow independent decay, two collective channels follow the equal-phase channel, and pairwise decay follows the opposite-phase channel. The decay rates show the split (Fig. 2d). Without protected operators the slowest rate grows in proportion to γ and the memory lifetime shrinks. With dark states the slow rates fall as $1/\gamma$ and the lifetime grows.

Why does stronger collective decay prolong the memory? In the two-spin picture of Fig. 1c, the dark state $|-\rangle \propto |10\rangle - |01\rangle$ can decay only through the bright state $|+\rangle \propto |10\rangle + |01\rangle$, and the Zeno effect [24, 41] suppresses this escape. Its rate falls as Ω^2/γ, where Ω couples $|+\rangle$ and $|-\rangle$, just as the slow mode of an overdamped oscillator settles more slowly the stronger the friction.

This protection follows from the strong-loss expansion at fixed system size. At strong loss the reservoir follows its Hamiltonian restricted to the dark states, which turns the stored state but cannot make it decay. Decay enters only at the next order, when the Hamiltonian briefly lifts a dark state into a bright one [25, 27]. One step is then a rotation plus a loss that falls as $1/\gamma$ (Eq. (A2)): a nearly orthogonal map with a small but finite decay rate, the classic recipe for a long memory in a network [31]. Calculations for three to six spins show the same decay out of the dark states, falling as $1/\gamma$ with different prefactors (Appendix E.1).

Two tests show that the dark states hold the memory (Appendix E.1). A model of the ten dark states alone, with no fitted parameter, reproduces the recall of the full reservoir at strong loss. Replacing the protected components of the state by their time averages after every step, which removes their dependence on the inputs, destroys all memory beyond a few steps, while the same operation on a randomly rotated set of equal size leaves most of it. Residual local noise caps this lifetime: for the equal-phase channel, our estimate requires a ratio of collective to local loss rates that grows quadratically with the target recall time (Appendix E.5).

Dark states thus protect stored information against decay. With the opposite-phase channel, however, the dark states are just as protected, yet the capacity returns only to its weak-loss level (Fig. 2c). The reason is the ongoing input.

3.2 New inputs mix stored information

Each new input turns the stored state slightly, by an angle that depends on the input (Fig. 1d), so to an old input every newer input acts as noise. Over

the memory lifetime the stored direction takes a random walk of small turns $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. Once the spread reaches about a radian, the record of an old input survives only multiplied by the inputs that followed it, and a linear readout cannot turn such products back into single inputs.

A stored contribution can therefore depend on one past input or on several jointly. We count the distinct past inputs that it depends on and call this number D. A function of u_{t-3} alone has $D = 1$, even if it is nonlinear in u_{t-3}, and $u_{t-3}u_{t-5}$ has $D = 2$. The information processing capacity [28, 42] also counts how well the readout reconstructs nonlinear functions of past inputs, and sorting its contributions by D separates information about single inputs from information about several jointly (Appendix A.4). The disturbance caused by one input is measured by its mixing fraction s, between 0 and 1: the part of a stored signal that one input turns differently from the average input. For small turns it grows as the square of the turn. The law concerns the covariance of the state, how its components fluctuate together across input histories, and assumes that one pattern of these fluctuations, a single slow direction of the covariance, outlives all others, even when many slow components of the state share it. Our main theorem then says that a stored contribution involves on average $s\,t_{\mathrm{mem}}$ inputs, the disturbance per input times the memory lifetime:

Theorem 1 (Mixing law, made precise in Appendix B)**.** Let a reservoir be reset with probability p per step, so that $t_{\mathrm{mem}} \sim 1/(2p)$ as $p \to 0$, and otherwise turned by input-dependent unitaries that mix the stored state, leave a single slow direction of the stored covariance, and write a signal (conditions in Appendix B). Pick a stored contribution with probability equal to its share of the capacity, and let D be the number of distinct inputs it depends on. As $p \to 0$, D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$

The reason is a race: in each step a fraction s of the stored signal picks up one more input, while a fraction $1/t_{\mathrm{mem}}$ of it is lost, so the number of inputs follows a geometric distribution, which becomes exponential for long memories. The same covariance condition extends the law to physical channels without a reset, with the slow decay in place of the reset (Appendix B.5), while several slow covariance directions keep distinct mixing scales (Appendix B.4). Nothing in the law is quantum: classical state-affine reservoirs, whose state is updated by an input-dependent linear map plus an offset [43], obey it under the same conditions. Dissipation enters through its two factors, the memory lifetime and the disturbance per input.

Exact covariance calculations, Eq. (A3), support the law in the five-spin reservoir (Fig. 3c). At the full input amplitude one input disturbs the stored state only slightly, $s \approx 0.02$, and the mean of D grows with the memory lifetime as $s\,t_{\mathrm{mem}}$ plus an offset of one to two inputs, as Theorem 1 allows (Appendix E.3). At a fixed input strength, a longer memory is therefore a more mixed one. How must the input strength change with the memory lifetime so that individual inputs remain readable?

3.3 Matching input strength to memory lifetime

The mixing law suggests weakening each new input as the memory lifetime grows. Because the disturbance s grows as the square of the turn, keeping $s\,t_{\mathrm{mem}}$ fixed requires an input amplitude proportional to $1/\sqrt{t_{\mathrm{mem}}}$. We call this gentle writing. A gentler input would write too little: one qubit makes the trade-off exact (Fig. 3d, Appendix C). Its stored signal is largest when the squared turn per input, accumulated over one relaxation time, is of order one, and only a turn that falls as $1/\sqrt{t_{\mathrm{mem}}}$ keeps the signal finite as the memory grows.

The single-input recall time shows the same balance. Its rate adds the decay of the stored signal and a mixing rate that grows as η^2 (Appendix A.3). The decay grows as γ with independent decay and falls as $1/\gamma$ with the opposite-phase channel, so $\eta^2 t_{\mathrm{f}}$ should depend on loss and input only through γ/η^2 and through $\gamma\eta^2$, respectively. Both collapses hold for four input amplitudes (Fig. 3a).

Gentle writing more than doubles the linear memory capacity. In the five-spin reservoir the amplitude falls as $1/\sqrt{\gamma}$ for the opposite-phase channel, whose lifetime grows with the loss, and as $\sqrt{\gamma}$ for independent decay, whose lifetime grows

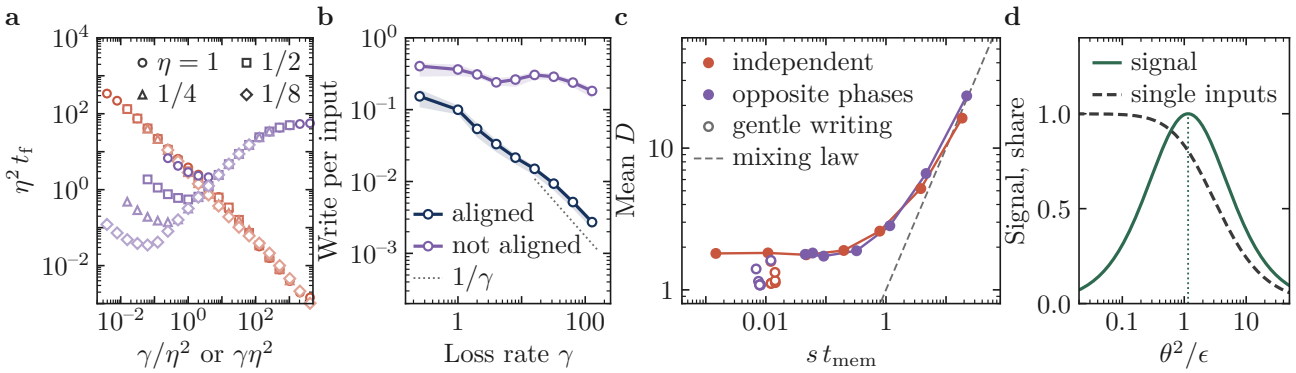


Figure 3: **New inputs mix stored information, and gentle writing limits the mixing. a** Single-input recall time for four input amplitudes η, rescaled as the sum of decay and mixing predicts: for each channel all amplitudes fall on one curve. **b** How strongly one input writes into the dark states of the equal-phase channel: an input aligned with the channel, the same on every spin, fades as $1/\gamma$, while a pattern that sums to zero over the spins keeps writing. **c** Mean number D of inputs mixed into a stored contribution of the five-spin reservoir, against the mixing law: at the full input strength it follows $s\,t_{\mathrm{mem}}$, and gentle writing keeps it near one. **d** One qubit, turned by θ per input and relaxing at the rate ϵ: the stored signal (solid, relative to its maximum) peaks at an intermediate θ^2/ϵ, the squared turn accumulated over one relaxation time (dotted), where single inputs (dashed) still carry 80% of the capacity.

as the loss weakens (Fig. 2c, circles). With exact readout the capacity rises from about 15 to 35 units for both channels (Table 3). Since the capacities of all orders add up to the 45 features [28], the gain moves capacity from combinations of inputs to individual inputs. A long memory therefore needs no dark states: weak independent decay with gentle writing reaches one too. What dark states add is a memory that stays long under strong dissipation and grows with the loss.

The phases of the collective channel decide whether an input reaches the protected memory at all. Spin i couples to the shared channel with the sign $c_i = \pm 1$, and the dark states are annihilated by the collective lowering operator $\sum_i c_i \sigma_i^-$. We call an input field aligned with the channel if its strength on each spin is proportional to c_i. An aligned field $\sum_i c_i X_i$ is the lowering operator plus its adjoint, so it cannot couple two dark states. It writes into them only through brief excursions into bright states, with a strength that falls as $1/\gamma$, while the non-aligned part of any other field writes into them directly (Fig. 3b, Appendix E.2). The uniform input of Eq. (1) is aligned with the equal-phase channel, so it writes gently by itself, and this channel comes close to the gain of gentle writing at the full amplitude. With the opposite-phase channel it is not aligned and must be weakened. This automatic gentle writing relies on exact alignment: a phase error opens a direct coupling within the dark subspace (Appendix E.2).

For three spins with the opposite-phase channel we prove both regimes (Appendix D). At a fixed amplitude single inputs lose their share of the capacity, while with an amplitude $\eta_0/\sqrt{\gamma}$ they keep it and are recalled ever further back. The record is written into coherences between the energy levels of the Hamiltonian restricted to the dark states. Coherence shapes the performance of spin reservoirs [44], and erasing these coherences after every input collapses the stored signal as $1/\gamma^2$. A gentler input, however, writes a fainter record. With exact readout this costs nothing. With measurements it does.

3.4 Measurement limits recoverable memory

Exact expectation values remove the cost of resolving weak signals. A memory whose input strength is matched to its lifetime grows longer, not larger: it cannot exceed the 45 features, so each delay is recalled more faintly.

For projective readout, each feature is estimated from repeated preparations of the reservoir. Each preparation replays a finite window of inputs, which grows with the memory lifetime, before it is measured [12]. We therefore count the budget in reservoir steps, preparations times replayed steps summed over the measurement settings, per output step, and simulate the replay (Appendix A.5).

Under this projective-readout protocol, weak independent decay with gentle writing recovers

the most memory at a fixed budget (Fig. 4a). At a budget of 10^9 reservoir steps per output step it keeps 34 units, nearly all of its exact capacity, against 21 for the opposite-phase channel at its best sampled loss rate near $\gamma = 64$. The dark-state memory lasts as long as that of weak independent decay, but its signal is about three times fainter. The equal-phase channel, whose exact capacity grows with the loss, now loses capacity as the loss grows.

The size of the signal is set by the state that the dissipation leaves behind. An input acts on the stationary state ρ_* through the commutator with its Hamiltonian, and only the part of this change that lies on the slow modes is remembered. We call its size the write strength. For this comparison we match the decay-only lifetime t_{leak}, the lifetime with the input held at its mean value, at 32 input steps (Appendix A.5). There the feature variance of the seven channels that write follows the write strength over four orders of magnitude (Fig. 4b). Collective decay with equal phases, through one channel or two, leaves a polarized state that the uniform input moves only through bright states, and it stores almost nothing. The opposite-phase channel leaves a similarly polarized state, but the input is not aligned with this channel and moves it directly (Sec. 3.3), so it stores far more. Independent and pairwise decay cool the spins into a state that the input moves directly, and they store the most. Consistent with earlier work on non-unital damping [7, 45], cooling helps here because the input can move the relaxed state along long-lived modes.

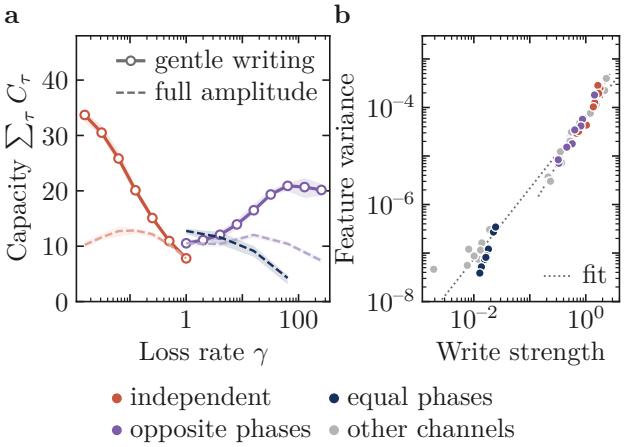


Figure 4: **Measurement limits recoverable memory.** **a** Recoverable memory capacity at a budget of 10^9 reservoir steps per output step: weak independent decay with gentle writing recovers the most. **b** The signal, the mean variance of the exact features over the inputs, grows with the write strength, how strongly an input changes the slow modes of the stationary state, for all channels at a matched decay-only lifetime.

4 Conclusion

A useful quantum reservoir must retain a readable record of the past while continuing to accept new information. We have shown that the structure of dissipation can fundamentally change how this record survives. Independent decay shortens memory as loss grows, whereas strong collective decay can prolong it by protecting information in dark states through the quantum Zeno effect. The environment therefore provides a physical means of designing the memory of a driven quantum system.

Longer storage also leaves more time for subsequent inputs to mix with the stored information. Our mixing law makes this accumulation quantitative. It identifies a practical principle: new information should be written more gently as memory grows, so that earlier inputs remain individually recoverable. In the spin reservoirs studied, this strategy more than doubles the capacity to recall past inputs with exact readout.

Measurement adds a further requirement: the stored information must produce a signal strong enough to resolve. Dark-state memories can remain protected for long times while becoming difficult to read. At a fixed measurement budget, weak independent decay recovers the most memory in our comparison because its signals are stronger. The resources needed to retrieve information are therefore part of the design of the memory itself.

Together, these results connect the protection of information stored in quantum dynamics to the practical demands of processing a continuous stream of inputs. They establish a quantitative design principle: match the memory lifetime to the disturbance caused by new inputs, and match the stored signal to the available measurement resources. Engineering the environment, the input, and the readout together provides a route from long-lived quantum dynamics to memory that remains useful during continued computation.

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Data availability

The code, the data behind every figure and the interval certificates of the three-spin limits are available at https://github.com/eybmits/what-a-dissipative-quantum-system-remembers, and the certificates are also attached to this PDF.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing, I.F. and J.W. to methodology, interpretation and editing, and C.L.-P. and J.S. to supervision and conceptual development.

A Methods

This appendix specifies the five-spin simulations and defines the quantities that the main text uses.

A.1 Reservoir, readout and capacity

The Hamiltonian of Eq. (1) uses $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$, so the transverse field runs from h to $2h$. Each input u is drawn uniformly from $[0, 1]$ and held for a step of length $\Delta t = 1/2$. We integrate Eq. (2) in the 1024-dimensional basis of Pauli operators and evaluate the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ by Chebyshev interpolation in u at ten nodes. Each reservoir draws its couplings and its input stream from one seed, every channel uses the same eight seeds, and runs with the input on one spin use the first four. After a washout of $\max(800, 60\gamma)$ steps, at least four times the single-input recall time, we record the 45 features, the expectation values of X_i, Y_i, Z_i and of X_iX_j, Y_iY_j, Z_iZ_j, for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. The linear memory capacity sums C_τ over delays 1 to 300. A delay without any recall reads about 1/1600 on the test inputs, so the sum has a floor of about 0.2 units. Figures show the mean of the eight reservoirs, with bands from the 16th to the 84th percentile.

Table 1: Decay channels at five spins, each with $\operatorname{Tr} K = 80\gamma$, and the number of operators that the dissipator leaves unchanged. The operators $\sigma_i^{\mp}$ lower and raise spin i, with $i < j$ for pairwise decay, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

channel	jump operators	protected operators
independent	$\sqrt{\gamma}\,\sigma_i^-$	1
equal phases	$\sqrt{\gamma}\sum_i \sigma_i^-$	100
opposite phases	$\sqrt{\gamma}\sum_i c_i\sigma_i^-$	100
two channels	$\sqrt{\gamma}\sum_{i\le 3}\sigma_i^-$, $\sqrt{\gamma}\sum_{i>3}\sigma_i^-$	36
pairwise	$\sqrt{\gamma}\,\sigma_i^-\sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\,\sigma_i^-$, $\sqrt{0.3\gamma}\,\sigma_i^-\sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\,\sigma_i^-$, $\sqrt{0.3\gamma/1.3}\,\sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\,Z_i$	32

A.2 Decay channels and equal coupling

Table 1 lists the jump operators of the eight decay channels. Their weights give every channel the same total coupling $\operatorname{Tr} K = N2^{N-1}\gamma = 80\gamma$, which fixes the mean decay rate. For a traceless jump L the superoperator trace of its dissipator is $-2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless, so the 4^N eigenvalues λ of the generator obey

$$\frac{1}{4^N}\sum \operatorname{Re}\lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{A1}$$

for every channel and every Hamiltonian. The protected operators form the kernel of the dissipator. Their number is not the number of dark states: the ten dark states of a collective channel carry $10 \times 10 = 100$ protected operators.

A.3 Slow modes, single-input recall time and the strong-loss generator

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues λ_j give decay rates $\kappa_j = -\ln|\lambda_j|/\Delta t$. If the dissipator protects n operators, the slow

modes are the $n-1$ slowest decaying ones besides the stationary state, and the single-input recall time is $t_{\rm f} = 1/(\Delta t\,\bar\kappa)$, with $\bar\kappa$ their mean rate. Without protected operators we use the slowest mode. The averaged step carries the part of a stored signal that still depends on a single input, so its mean rate adds decay and mixing exactly (Appendix E.2). Figure 2d shows instead the decay rates $-\operatorname{Re}\lambda$ of the generator at the mean input, $u = 1/2$. With a collective channel we call the $n-1$ slowest of them dark and all faster ones bright.

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D}+\mathcal{P})^{-1} - \mathcal{P}$ its inverse on the decaying operators. At strong loss, eliminating the decaying operators to second order [25, 27] gives the generator of the protected coordinates,

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\,\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \qquad \text{(A2)}$$

with $\mathcal{H} = -i[H,\cdot\,]$. The first term is the Hamiltonian restricted to the protected operators, which turns them, and the second is the loss through brief excursions into bright states.

Gentle writing uses the amplitude $\eta = \min(1,\gamma^{-1/2})$ of Sec. 2 for the opposite-phase channel and $\eta = \min(1,\gamma^{1/2})$ for independent decay.

A.4 Number of inputs mixed

The order D of a stored signal is the number of distinct past inputs it depends on (Sec. 3.2). Precisely, for independent inputs, every function of the past inputs with finite variance splits uniquely into orthogonal parts, each of which depends on a fixed finite set of inputs. The order of a part is the size of its set, not a polynomial degree. For signs $u_t = \pm 1$ the parts are the products $\prod_{j\in J} u_{t-j}$, and for continuous inputs one uses orthogonal polynomials in each input. We collect the traceless coefficients of the state in an orthonormal operator basis in a vector r, on which quantum channels act affinely [46, 47], $r_t = A(u_t)r_{t-1} + b(u_t)$. All means and covariances are taken over input histories. Let $\bar A = \mathbb{E}A$ be the mean step, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, the change that one input writes, with covariance V. With $\mathcal{B}X = \bar A X \bar A^{\mathsf T}$, $\mathcal{T}X = \mathbb{E}[AXA^{\mathsf T}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1-\mathcal{B})^{-1}V, \qquad G_{k+1} = (1-\mathcal{B})^{-1}\mathcal{K}G_k. \qquad \text{(A3)}$$

The innovation adds the current input alone. The mean step carries a stored product along without adding an input. The fluctuating part $\mathcal{K}$ of the step multiplies it by the current input, which raises its order by one. The G_k add up to the stationary covariance Σ. A readout of features $y = Mr$ has the capacity $\operatorname{Tr}[(M\Sigma M^{\mathsf T})^{+} M G_k M^{\mathsf T}]$ of order k, and these capacities add up to the rank of the readout [28]. Divided by their sum, they give the share of each order, the distribution of D in Theorem 1. For one observable with coordinates a the share of order k is $a^{\mathsf T}G_k a / a^{\mathsf T}\Sigma a$, and the share of single past inputs is $a^{\mathsf T}\bar A G_1 \bar A^{\mathsf T} a / a^{\mathsf T}\Sigma a$.

The memory lifetime and the mixing fraction come from the slowest mode of the covariance step $\mathcal{T}$: its eigenvalue is $1 - 1/t_{\rm mem}$, and with its right eigenvector Q and left eigenvector ℓ the mixing fraction is $s = \ell(\mathcal{K}Q)/\ell(Q)$, as in Appendix B.5. For five spins we evaluate Eq. (A3) on the 1023 traceless Pauli coordinates (Appendix E.3).

A.5 Finite sampling and the measurement budget

A quantum measurement gives a random outcome, so every feature is an average over n preparations per measurement setting and step. We use three settings, with all spins read along X, Y or Z. For each step we form the outcome distributions of the three settings and draw n preparations from each. The readout is fitted as above, with a ridge chosen from fourteen values between 0 and 10^4 on the validation inputs. In Sec. 3.4 the distributions come from the replay: each output step replays its last τ_{90} inputs from $|00000\rangle$, where the first τ_{90} delays hold 90% of the exact recall. We count the budget B in reservoir steps, preparations times replayed steps summed over the three settings, per output step, so that $n = B/(3\tau_{90})$, and we also run $n = 10^3$ to $10^{8.5}$ in half decades. Only Fig. 4b uses the state with the full history. Its matched decay-only lifetime is the leak time $t_{\rm leak} = 1/(\Delta t\,\bar\kappa_0)$ of 32 input steps, where $\bar\kappa_0$ is the mean decay rate of the slow modes of the generator at the mean input, without the mixing by the inputs (Appendix E.4).

B The mixing law

This appendix proves the mixing law of Sec. 3.2 and extends it to physical channels without a reset, which Appendix D needs.

B.1 Setting and statement

We describe the reservoir by the n traceless coordinates r of its state in an orthonormal basis of operators, the Bloch vector for a qubit. A unitary acts on r as a rotation. In each step the reservoir is reset to a fixed state with a small probability p and is otherwise rotated by $O(u)$:

$$r_t = (1-p)\,O(u_t)\,r_{t-1} + p\,b. \tag{B1}$$

Here b represents the reset state. Equation (B1) averages over whether the reset occurred, so the only remaining randomness is the input history, with independent and identically distributed inputs u_t. A stored signal loses a fraction $1-(1-p)^2 = 2p-p^2$ of its variance per step, so its memory lifetime, defined by the slow eigenvalue $1-1/t_{\mathrm{mem}}$ of the covariance step (Appendix A.4), is $t_{\mathrm{mem}} = 1/(2p-p^2)$, or $1/(2p)$ to leading order. Three conditions make the law hold:

(1) *The average step forgets.* The average rotation $A = \mathbb{E}O$ shrinks every vector: its spectral radius is below one.

(2) *The inputs mix.* The only covariance matrices that the random rotations leave unchanged on average, $\mathbb{E}[OXO^{\mathsf{T}}] = X$, are multiples of the identity.

(3) *The input writes.* One input changes the state: $\mathbb{E}\|(O-A)\mu\|^2 > 0$ with $\mu = (1-A)^{-1}b$.

The first condition means that only products of inputs can be stored for long, the second that a stored signal spreads evenly over all directions, and the third that there is something to store. The mixing fraction

$$s = 1 - \frac{\|A\|_F^2}{n} \tag{B2}$$

says how strongly one input turns the stored state. For a unit vector x of random orientation, $\mathbb{E}\|(O-A)x\|^2 = s$: one input turns a fraction s of a stored signal away from the average rotation. The fraction lies between 0 and 1, and the first condition makes it positive. If the inputs $u = \pm 1$ rotate by $O = e^{u\theta G}$ with a fixed generator G, whose eigenvalues are $i\lambda_1, \dots, i\lambda_n$, the average rotation is $\cosh \theta G$ and $s = \frac{1}{n}\sum_j \sin^2 \theta\lambda_j$, which grows as θ^2 for small turns.

> **Mixing law, precise form.** Assume conditions (1)–(3) and take any fixed linear readout. Pick a stored contribution with probability equal to its share of the readout's capacity, and let D be its order, the number of distinct inputs it depends on. As $p \to 0$, the ratio D/t_{mem} becomes exponentially distributed with mean s, and
>
> $$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$
>
> Equivalently, products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $e^{-x/s}$ of the capacity.

B.2 The idea: a race

In the race of Sec. 3.2, on average $1/s$ steps pass before the next input joins a product, and meanwhile it loses a fraction $1/(s\,t_{\mathrm{mem}})$ of its weight, so products of more than K inputs carry about

$$\Big(1 - \frac{1}{s\,t_{\mathrm{mem}}}\Big)^K \approx e^{-K/(s\,t_{\mathrm{mem}})},$$

an exponential law with mean $s\,t_{\mathrm{mem}}$. The proof makes this bookkeeping exact.

B.3 Proof

The proof works with covariance matrices. Appendix A.4 splits the stationary covariance Σ of the state into the parts G_k carried by products of k inputs and gives their recursion, Eq. (A3): the mean step $\mathcal{B}$ carries a stored product along, and the fluctuating part $\mathcal{K}$ of the step adds one input to it. In the reset model the mean step is $\mathcal{B}X = (1-p)^2 AXA^{\mathsf{T}}$, the full step $\mathcal{T}X = (1-p)^2\,\mathbb{E}[OXO^{\mathsf{T}}]$, and $\mathcal{K} = \mathcal{T} - \mathcal{B}$. We write $\mathcal{B}_0$, $\mathcal{T}_0$ and $\mathcal{K}_0$ for the same maps at $p = 0$.

Step 1: the stored signal is small and spread evenly. The stationary mean is $m = p\,[1-(1-p)A]^{-1}b = p\mu + O(p^2)$, so one input writes the change $(1-p)(O-A)m$, of size p. Its covariance V has the trace $p^2 v + O(p^3)$ with $v = \mathbb{E}\|(O-A)\mu\|^2$, which is positive by condition (3). The random rotations preserve the trace of a covariance and, by condition (2), leave only the identity unchanged. Since they cannot enlarge a covariance,

the identity is a simple fixed point with the projector $X \mapsto (\operatorname{Tr} X/n)\,1$. On every other direction $(1-\mathcal{T})^{-1}$ stays bounded as $p \to 0$, while on the identity it equals $1/[1-(1-p)^2] \approx 1/(2p)$. The written covariance therefore piles up along the identity:

$$\Sigma = (1-\mathcal{T})^{-1} V = \frac{pv}{2n}\,1 + O(p^2). \tag{B3}$$

Step 2: few inputs carry almost nothing. The mean step forgets by condition (1), so $(1-\mathcal{B})^{-1}$ stays bounded as $p \to 0$. The recursion then makes $G_1 = (1-\mathcal{B})^{-1}V$ of order p^2, and the same holds for every G_k with fixed k. Measured against Σ, which is of order p and has full rank, any fixed set of orders carries a share of order p of the capacity of any readout. A long memory consists of products of many inputs.

Step 3: counting the inputs. Call $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ the order transfer, so that $G_{k+1} = \mathcal{F}G_k$. At $p = 0$ it leaves the identity unchanged, because $\mathcal{K}_0 1 = 1 - AA^{\mathsf{T}} = (1-\mathcal{B}_0)1$. Its matching left eigenvector is the weight

$$\ell(X) = \operatorname{Tr}[(1 - A^{\mathsf{T}}A)X], \qquad \ell(1) = ns, \tag{B4}$$

since $\ell(\mathcal{F}_0 X) = \operatorname{Tr}\mathcal{K}_0 X = \ell(X)$ when the rotations preserve the trace. For small p the eigenvalue one moves to

$$1 - \frac{2}{s}\,p + O(p^2)\,: \tag{B5}$$

each step up in order costs a fraction $2p/s = 1/(s\,t_{\mathrm{mem}})$ of the weight, as in the race. To see this, differentiate $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $p = 0$ and apply it to the identity, which gives $(1-\mathcal{B}_0)\mathcal{F}_0' 1 = -2\cdot 1$. Its trace is the weight $\ell(\mathcal{F}_0' 1) = -2n$, and dividing by $\ell(1) = ns$ gives the rate $2/s$.

To turn the eigenvalue into a distribution, weigh order k with $e^{-\lambda p k}$. Summing the recursion gives $\sum_k z^k G_k = z(1-z\mathcal{F})^{-1}G_1$, and as $z = e^{-\lambda p}$ approaches one, only the eigenvalue of Eq. (B5) makes the inverse large. With $\ell(G_1) = p^2 v + O(p^3)$ and $\beta = 2/s$,

$$\sum_{k\geq 1} e^{-\lambda p k} G_k = \frac{pv}{2n}\,\frac{\beta}{\lambda+\beta}\,1 + O(p^2). \tag{B6}$$

Divided by Eq. (B3), this is $\beta/(\lambda+\beta)$, the Laplace transform of an exponential distribution of pD with rate β. So $D/t_{\mathrm{mem}} = 2pD\,[1 + O(p)]$ is exponential with mean s in the limit. Because both covariances are multiples of the identity to leading order, the same holds for the share of the capacity of any readout. Finally, $\sum_k k G_k = (1-\mathcal{F})^{-1}\Sigma = \Sigma/(\beta p) + O(p)$ gives $\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1)$.

Two technical points. The eigenvalue one of $\mathcal{F}_0$ is simple: a fixed point of $\mathcal{F}_0$ satisfies $\mathcal{T}_0 X = X$ and is therefore a multiple of the identity, and $\ell(1) > 0$ rules out a Jordan chain. For $p > 0$ every eigenvalue of $\mathcal{F}$ lies inside the unit circle, by a Perron–Frobenius argument: $\mathcal{F}$ maps positive matrices to positive matrices, and so does $(1-\mathcal{F})^{-1} = 1 + (1-\mathcal{T})^{-1}\mathcal{K}$. The eigenvalues of $\mathcal{F}_0$ therefore lie in the closed unit disk, only the simple one equals one, and only the eigenvalue of Eq. (B5) makes the sum diverge.

B.4 When the law needs one slow direction

Condition (2) matters. Let a classical flag, set by the reset state and never changed, select one of two qubit rotations about Z, by $\pm\pi/6$ or by $\pm\pi/3$, and let the reset state put the flag in either value with probability $\frac{1}{2}$ and the qubit along $+X$ in both. Each sector obeys the law with its own mixing fraction, $1/4$ and $3/4$, and a single Y readout sees their mixture, weighted by the variance each sector stores. The gentler sector stores more, so as $p \to 0$ products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $w\,e^{-4x} + (1-w)\,e^{-4x/3}$ with $w = 0.82$, which exact moments reproduce. One observable then shows two order scales.

B.5 Physical channels without a reset

A physical channel does not reset to a prescribed state, but the proof used the reset only through three facts: one covariance direction is slow, it leaks at a small rate, and the mean step forgets everything else. The same three steps therefore prove the law for general channels.

Setting. One step is $r_t = A_u(\epsilon) r_{t-1} + b_u(\epsilon)$ on the coordinates that the inputs can reach, with a small leak ϵ, and $\mathcal{B}$, $\mathcal{T}$ and $\mathcal{K}$ are formed from A_u as before. We assume:

(i) A_u and b_u depend smoothly on ϵ in mean square, and b_u vanishes at $\epsilon = 0$.

(ii) The mean step forgets at $\epsilon = 0$: $\mathbb{E}A_u(0)$ has spectral radius below one.

(iii) The channel is mean-square stable for small $\epsilon > 0$: the covariance step $\mathcal{T}$ has spectral radius below one.

(iv) One covariance direction is slow: $\mathcal{T}$ has a simple eigenvalue $1 - \kappa\epsilon + O(\epsilon^2)$ with $\kappa > 0$ and no other eigenvalue near one. At $\epsilon = 0$ its right eigenvector is a positive matrix Q, and its left eigenvector is a weight ℓ that is nonnegative on positive matrices, with $\ell(Q) = 1$.

(v) The input writes along it: the change written by one input, of order ϵ, has covariance $\epsilon^2 V_* + O(\epsilon^3)$ with $\ell(V_*) > 0$.

The reset model is the case $\epsilon = p$, $\kappa = 2$, $Q = 1$ and $\ell = \mathrm{Tr}\,/n$. Condition (iv) concerns the covariance that the random steps carry, not the number of slow modes: at a fixed amplitude the inputs turn the 99 slow dark modes of the opposite-phase channel differently in each step and spread a stored signal over all of them, so that one covariance direction stays slow. Conditions (ii) and (iii) say that the reservoir forgets its starting state on average and in mean square, (iv) that the random steps spread a stored signal over the slow modes faster than it leaks, and (v) that the surviving direction is written.

Mixing law without a reset. Assume (i)–(v) and take a fixed readout $y = Mr$ whose projected slow covariance MQM^T has full rank. Then the mixing fraction $s = \ell(\mathcal{K}_0 Q)$ is positive and at most one, and ϵD becomes exponentially distributed with rate κ/s. The mean order is $s/(\kappa\epsilon) + O(1)$: the mixing fraction times the memory lifetime $1/(\kappa\epsilon)$.

Proof. Steps 1 and 2 carry over. The stationary mean is of order ϵ by (i) and (ii), and the slow eigenvalue piles the written covariance up along Q,

$$\Sigma = \epsilon \, \frac{\ell(V_*)}{\kappa} \, Q + O(\epsilon^2),$$

and every fixed order carries a covariance of order ϵ^2. Here the readout condition enters: the shares of the orders in the capacity of the readout follow from Σ only if MQM^T has full rank. For one observable a this requires $a^\mathsf{T} Q a > 0$. In step 3, $\mathcal{F}_0$ has the same fixed points as $\mathcal{T}_0$, so Q spans them, and $\ell \circ (1 - \mathcal{B}_0)$ is its left eigenvector, with value $\ell(\mathcal{K}_0 Q) = s$ on Q. Differentiating $(1 - \mathcal{B})\mathcal{F} = \mathcal{K}$ at $\epsilon = 0$ and applying it to Q gives $(1 - \mathcal{B}_0)\mathcal{F}_0' Q = \mathcal{T}_0' Q$, whose weight $\ell(\mathcal{T}_0' Q) = -\kappa$ is the rate of the slow eigenvalue of $\mathcal{T}$. The order eigenvalue is therefore $1 - (\kappa/s)\epsilon + O(\epsilon^2)$, and the Laplace transform follows as before. It remains to show that s is positive, which also makes the eigenvalue simple. Because $\mathcal{K}_0$ maps positive matrices to positive matrices and ℓ is nonnegative on them, $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every positive Y, so $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 Q) \le s \le 1$ for every j. If s vanished, the series $\mathcal{F}_0 = \sum_j \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 Q) = 0$, which contradicts $\mathcal{F}_0 Q = Q$ and $\ell(Q) = 1$.

The spin reservoirs. For three spins the certificates of Appendix D check these conditions. For five spins we check the law instead of the conditions: Eq. (A3) gives the order distribution exactly, and it follows the law (below).

B.6 Numerical check

Figure 5 compares the law with exact moments. Beyond the reservoir shown, we tested 24 reservoirs and readouts in dimensions two to four: two families of three random unitaries, two reset states (a basis state and a random pure state) and two readouts (all coordinates or one random direction). The largest deviation from the law shrinks in proportion to p, from 0.08 at $p = 3\%$ to 0.003 at $p = 0.1\%$. The five-spin reservoir follows the law as well (Fig. 3c): at the largest memories computed, $s\,t_{\mathrm{mem}} \approx 20$, products of more than K inputs carry the share $e^{-K/(s\,t_{\mathrm{mem}})}$ of the capacity to within 0.14.

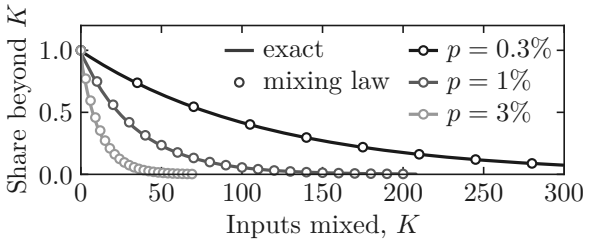


Figure 5: **The mixing law.** Share of the capacity carried by products of more than K inputs, for one qubit that is reset with a small probability p in each step: exact, and from the mixing law of Theorem 1.

C One qubit in closed form

This appendix solves the qubit of Sec. 3.3 exactly.

The model. The qubit relaxes towards $+X$ at a rate ϵ, through the jump $\sqrt{\epsilon}\,|+x\rangle\langle -x|$, and the input $u = \pm 1$ turns it about Z by $u\theta$ per unit time. Each input lasts one unit of time, and the qubit is read along Y. In the Bloch plane,

$$\dot{x} = -\epsilon(x-1) - u\theta y, \qquad \dot{y} = u\theta x - \tfrac{1}{2}\epsilon y. \quad \text{(C1)}$$

Over one step this is a linear map of (x, y) plus an offset. For $u = +1$ the map has the form $\left(\begin{smallmatrix} a & -b \\ b & d \end{smallmatrix}\right)$ and the offset is (f, g). For $u = -1$ the signs of b and g flip. The input-averaged step is therefore diagonal, the stationary mean lies on the X axis at $m = f/(1-a)$, and each input writes $u\,v$, with $v = bm + g$, along the measured axis Y.

How the orders arise. The averaged step keeps the two axes apart, while the input-dependent part swaps them. A signal written into Y must pass through X and back to pick up more inputs, so Y carries only odd orders. By the recursion (A3), covariance passes from Y to X with the factor $b^2/(1-a^2)$ and back with $b^2/(1-d^2)$. Call the product of the two factors χ. Order $2j+1$ then carries the share $(1-\chi)\chi^j$ of the capacity of Y, and

$$\mathbb{E}D = \frac{1+\chi}{1-\chi}, \qquad \operatorname{Var} Y = \frac{v^2}{(1-d^2)(1-\chi)}. \quad \text{(C2)}$$

Single past inputs carry the share $d^2(1-\chi)$.

Fixed turn. Let the relaxation slow down, $\epsilon \to 0$, at a fixed angle θ with $\sin\theta \neq 0$. Then $1-\chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$: the order grows as $1/\epsilon$, ϵD becomes exponentially distributed with rate $3/(2\sin^2\theta)$, and the share of single inputs vanishes, even for exact expectation values.

Gentle writing. If instead θ^2/ϵ approaches a constant ν, then to first order $a = 1-(1+\nu/2)\epsilon$ and $d = 1-(1+\nu)\epsilon/2$, with $b \approx \theta$ and $f \approx \epsilon$, and as $\epsilon \to 0$

$$\chi \to \frac{\nu^2}{(\nu+2)(\nu+1)}, \qquad \operatorname{Var} Y \to \frac{4\nu}{(\nu+2)(3\nu+2)}. \quad \text{(C3)}$$

The mean order stays finite, and single past inputs keep the share $1-\chi$. Their recall beyond delay τ is the fraction $d^{2\tau}$ of the total, so the range that holds 90% of it grows as $1/\epsilon$: $\epsilon\tau_{90} \to \ln 10/(1+\nu)$.

The best turn. The signal of Eq. (C3) is largest at $\nu = 2/\sqrt{3}$ (Fig. 3d). For $\nu \gg 1$ it vanishes as $4/(3\nu)$, because later inputs mix earlier ones into products. For $\nu \ll 1$ it vanishes as ν, because each input writes too little. Among power laws $\theta \propto \epsilon^\alpha$, only $\alpha = 1/2$ keeps a finite signal: the turn must fall as the square root of the relaxation rate, that is, as one over the square root of the memory lifetime. Direct propagation of the density matrix reproduces all these expressions.

D Three spins at strong loss

What we prove. Take three spins with a collective channel of opposite phases, Eq. (D1), the four Hamiltonians H1–H4 of Table 2, and every Hamiltonian whose three couplings lie in a small box around theirs. As γ grows:

- At a fixed input amplitude, the order of the memory of X_1, divided by γ, becomes exponentially distributed, and the signal in X_1 falls as $1/\gamma$ (boxes of half-width 1/32).
- With the amplitude scaled as $8/\sqrt{\gamma}$, single past inputs keep a finite share of the capacity of X_1, and the range of delays that holds 90% of their recall grows in proportion to γ (boxes of half-width 1/512).

Idea. At strong loss, one step of the full master equation is a rotation of the three dark states, up to corrections of order $\epsilon = 1/\gamma$. At a fixed amplitude the two signs of the input rotate the dark states differently. The stored covariance then has a single slow direction, and the mixing law without a reset applies (Appendix B.5). With gentle writing both signs give almost the same rotation. The signal is written on the scale of the leak and collects in six slow directions, which we follow explicitly. Both arguments end in finitely many strict inequalities, which a computer checks with rigorous error bounds.

Model and computations. The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij} X_i X_j + \tfrac{1}{2}\sum_i Z_i + \tfrac{3+\eta u}{4}\sum_i X_i \quad \text{(D1)}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ (fixed

Table 2: Strong-loss limits for X_1 in the four three-spin reservoirs. Fixed amplitude: rate of the exponential law of the order divided by γ (certified). Gentle writing: share of single past inputs and range of their recall divided by γ, evaluated at $\gamma = 2^{16}$.

	rate	single-input share	range/γ
H1	78.6	0.81	0.50
H2	91.7	0.29	0.18
H3	61.4	0.08	0.24
H4	132.5	0.14	0.50

amplitude) or $\eta = 8/\sqrt{\gamma}$ (gentle writing): the field of gentle writing (Sec. 2) with signs in place of uniform inputs, with the same mean and the same largest change $\eta/4$. The couplings of the four Hamiltonians H1–H4 are drawn from fixed seeds. For each loss rate we exponentiate the full generator for both signs, solve the stationary moment equations and evaluate Eq. (A3) for X_1 until 99.99% of its memory is accounted for. The range τ_{90} is the smallest delay beyond which less than a tenth of the recall of single inputs remains.

Both regimes. Figure 6 shows the two regimes of the box for H3. Gentler writing trades signal for readability, as for the qubit: at $\eta_0 = 8$ single inputs keep between 8% and 81% of the capacity of X_1 for the four Hamiltonians (Table 2), and for H3 this share rises to 67% at $\eta_0 = 2$, with a signal 25 times smaller.

One step at strong loss. The jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$ leaves three dark states: the state with no excitation and the two single-excitation states with amplitudes $(0, 1, -1)/\sqrt{2}$ and $(2, 1, 1)/\sqrt{6}$, which are orthogonal to the phases of the channel. It empties every other state, so at unit loss rate only the nine operators on the dark states do not decay. Let R embed these nine coordinates into the full operator space and $R^\flat$ project back along the decaying operators. Expanding one step in ϵ gives

$$E_u = R\, e^{\Delta t\, R^\flat \mathcal{H}_u R} R^\flat + \epsilon Z_u + O(\epsilon^2), \qquad \text{(D2)}$$

where $\mathcal{H}_u$ is the commutator with the Hamiltonian of input u. The first term rotates the dark states. The correction Z_u collects the brief excursions into bright states: one at the start of the step, one at its end, and the leak of Eq. (A2) in between.

Fixed amplitude. At $\eta = 1$ the dark rotations O_u of the two signs differ, and the computer checks the conditions of the mixing law without a reset on the eight traceless dark coordinates. The average rotation forgets. The identity is the only covariance that the random rotations leave unchanged. Three constants from the first-order terms of Eq. (D2) are positive: the strength with which an input writes, the rate κ at which the slow covariance leaks, and the mixing fraction $s_0 = 1 - \|\mathbb{E} O_u\|_F^2/8$. The law then gives an exponential distribution of the order divided by γ, with rate κ/s_0 (Table 2), and the variance of X_1 falls as ϵ. A change r of the couplings moves each dark rotation by at most $3r$, which lets the checks cover the boxes of half-width 1/32.

Gentle writing. At $\eta = 8\sqrt{\epsilon}$ both signs give the same leading rotation E, generated by the static part of the dark Hamiltonian. The input enters at order $\sqrt{\epsilon}$ and the leak at order ϵ. Expanding the step to this order gives the mean step $E + \epsilon E_1$ and the input-dependent part $u\sqrt{\epsilon}\,\Delta_1$.

Where the signal goes. The dark energy levels are the eigenstates of the static dark Hamiltonian. For each pair of them, the real and imaginary parts of their coherence span a plane, and E turns each of these three planes by its own angle, set by an energy gap. The six phase factors of these turns differ from each other and from one. So E leaves only the populations unchanged, and on them E_1 generates a slow population dynamics with a unique stationary state m_*. An input therefore writes $\pm\sqrt{\epsilon}\, v$ with $v = \Delta_1 m_*$, which lies in the coherence planes: a new input first turns a population into a coherence.

Six slow directions. The covariances that E leaves unchanged form a six-dimensional space: the three covariances of the two population coordinates and one isotropic covariance in each coherence plane. Over $1/\epsilon$ steps a covariance in this space changes under two generators, both certified stable: K_B and K_T, the first-order terms in ϵ of the mean and the full covariance step, $\mathcal{B}$ and $\mathcal{T}$, restricted to the six directions. K_B lets a stored signal decay, and K_T adds the random signs of later inputs, which turn a stored signal into products. With P_6 the projection onto the six directions, the single-input part of the stored covariance and the full covariance approach $W_* = -K_B^{-1} P_6(vv^\mathsf{T})$ and $\Sigma_* = -K_T^{-1} P_6(vv^\mathsf{T})$.

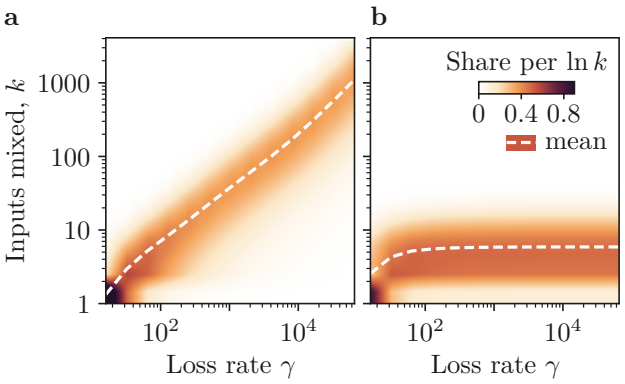


Figure 6: **Three spins, both regimes.** Share of the capacity of X_1 per logarithmic interval of k, the number of inputs combined (colour), against the loss rate, for the opposite-phase channel (Hamiltonian H3). At a fixed input strength the number of inputs mixed grows with the loss (a), while with gentle writing it stays small (b).

Limits. With a the coordinates of X_1, the share of single past inputs in the memory of X_1 tends to $a^\mathsf{T} W_* a / a^\mathsf{T} \Sigma_* a$. After t/ϵ steps the remaining single-input recall is the fraction $a^\mathsf{T} e^{tK_B} W_* a / a^\mathsf{T} W_* a$, which crosses 0.1 once, and the crossing is the range divided by γ. The certificates show $a^\mathsf{T} W_* a > 0$, and since $\Sigma_* \succeq W_*$ the share is positive. In the boxes of half-width $1/512$ the six angles stay separated and both generators stay stable, so the limits stay positive.

Why coherence carries the memory. Returning bright population to the dark subspace after every step leaves the leading terms of the scaled step unchanged, so it keeps the same limits. Dephasing in the dark energy basis, by contrast, removes the coherence planes into which v writes. Restricted to populations, the two signs then differ only at order $\epsilon^{3/2}$, because the response of a population at order $\sqrt{\epsilon}$ is a coherence, while the populations still follow the slow population dynamics. A new input thus writes a vector of size $\epsilon^{3/2}$, the slow dynamics sum its square over $1/\epsilon$ steps, and the stored signal is of order ϵ^2: the collapse as $1/\gamma^2$ of Sec. 3.3. Numerically, applying the loss-only projector $\mathcal{P}$ after every step, with or without dephasing in the energy basis of the projected Hamiltonian, the variance of X_1 stays between 10^{-4} and 2×10^{-3} from $\gamma = 2^8$ to 2^{18} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased. At a fixed input amplitude it falls as $1/\gamma$.

What the computer checks. The attached archive `spin_certificates.zip` contains the generators, the exact couplings and the interval programs. They enclose every quantity above in intervals with rigorous rounding (ball arithmetic of Arb, python-flint 0.9.0) and check each strict inequality of this appendix at 256 bits, and at 192 bits they reproduce the archived enclosures. A second program, sharing no routine with them, builds the full finite-rate maps and extrapolates to infinite loss. It agrees with the certified rates and limiting recalls to better than one part in a million.

E Numerical details and supporting checks

This appendix backs the main text, section by section, with what we checked and what is needed to reproduce it.

E.1 Memory lifetime and dark states (Sec. 3.1)

Figures smooth recall curves for display only: each delay is averaged with its neighbours, with weights that fall linearly to zero at a fifth of the delay, and no sum uses the smoothed values. The maps of Fig. 2a,b are computed at $\gamma = 0.1$, 0.25 and 2^{-1} to 2^6 in powers of two and interpolated by shifting each recall profile along the delay. Figure 2c stops the equal-phase channel and the two collective channels at $\gamma = 64$, where their recall at the full amplitude starts to reach past the 300 delays we sum. Summed over all delays with exact moments (Appendix A.4), their capacity keeps rising up to $\gamma = 256$. With gentle writing, delays beyond 300 add about 1% to the capacity of the opposite-phase channel.

Both tests of the dark states act on the 100 dark coordinates. For the dark-state model we write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the dark coordinates α, evolve α under Eq. (A2) and record the same 45 features of the full state $(R - \mathcal{S}\mathcal{H}(u)R/\gamma)\,\alpha$. The readout is trained on them exactly as for the full reservoir, and the model has no fitted parameter (Fig. 7a). For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar{r})$. The control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is measured at delays 1 to 100 with 800 training,

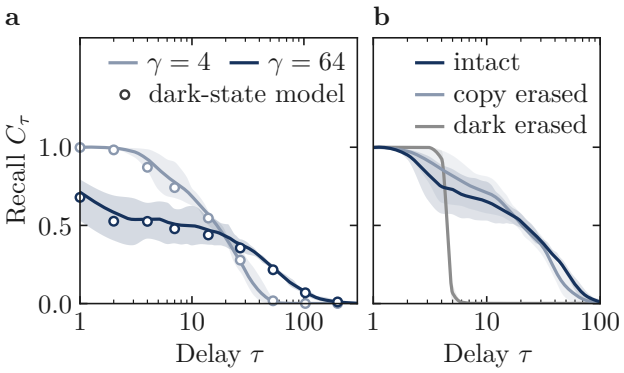


Figure 7: **Dark states hold the memory.** Equal-phase channel. **a** A model of the dark states alone, with no fitted parameter, reproduces the recall of the full reservoir (lines). **b** Replacing the dark coordinates by their time averages after every step destroys the memory, while the same operation on a randomly rotated set of equal size leaves most of it ($\gamma = 16$).

200 validation and 800 test inputs (Fig. 7b).

The decay out of the dark states falls exactly as $1/\gamma$: for the equal-phase channel $\gamma\bar{\kappa}$ approaches $\kappa_c = 2.945$ from $\gamma = 512$ on, so t_f grows in proportion to γ, while it levels off near 56 steps with the opposite-phase channel and near 110 with pairwise decay. The second-order calculation of Eq. (A2) gives the same κ_c at five spins and 1.4, 2.1 and 3.4 at three, four and six spins: at every size we computed, the decay falls as $1/\gamma$, with a prefactor that depends on the size.

E.2 Recall time and phases (Secs. 3.2 and 3.3)

We compute t_f at $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$, for γ from 2^{-8} to 2^6 with independent decay and from 2^{-2} to 2^{12} with the opposite-phase channel. The collapses of Fig. 3a do not depend on how the recall time is defined: with the slowest mode in place of the mean rate they hold to within 11% wherever the recall time exceeds four steps, and they hold as well with the input on one spin. Weakening the input moves the preferred loss rate: on one reservoir with exact moments, the sampled capacity peak of independent decay moves from $\gamma \approx 1/16$ at $\eta = 1$ to $1/64$ at $\eta = \frac{1}{8}$, and the peak capacity rises from 16 to 39 units.

The write per input of Fig. 3b is $2\|R^b[E(\frac{3}{4}) - E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1, \dots, 1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. The aligned pattern falls as $1/\gamma$, to 0.003 at $\gamma = 128$, while the other falls only slowly, to 0.18. The uniform input of Eq. (1) is aligned with the equal-phase channel, and its single-input recall time keeps growing in proportion to γ. It is not aligned with the opposite-phase channel, and its recall time levels off. The rule is sharp: a phase error δ couples dark states directly at order δ, so beyond $\gamma \sim 1/\delta$ the equal-phase channel behaves like the opposite-phase channel. The opposite-phase channel is the equal-phase channel with the field on spins 4 and 5 reversed, up to a relabelling of the random couplings (a turn of these spins by π about Z).

E.3 Inputs mixed and capacities (Secs. 3.2 and 3.3)

The distribution of the number of inputs mixed, Eq. (A3), uses six-point Gauss–Legendre quadrature for the expectations over the input and solves the equations $X - \bar{A}X\bar{A}^\mathsf{T} = C$ in the eigenbasis of $\bar{A}$. The shares of the 45 features are summed until the rest falls below 10^{-8}, or until successive orders shrink at a settled ratio, whose geometric tail we add. The linear recall follows from the same moments, $C_\tau = c_\tau^\mathsf{T}\Sigma_y^{-1}c_\tau/\mathrm{Var}\,u$ with c_τ the covariance of the features with $u_{t-\tau}$ and Σ_y their covariance, and reproduces the simulated capacities to a median of 2%. Three steps of inverse iteration on $\mathcal{T}$, from the input's covariance V and from the identity, give t_{mem}, Q and ℓ. Loss rates run in powers of four over the ranges of Fig. 3c, and with gentle writing to 16384 and 1/4096, on four reservoirs, two at the longest memory lifetimes. At the full amplitude the mean order of the opposite-phase channel rises from 1.8 at $\gamma = 1$ to 23 at $\gamma = 4096$, and with independent decay it stays within 2.2 of $s\,t_{\mathrm{mem}}$ up to $s\,t_{\mathrm{mem}} = 18$.

Table 3 collects the capacities. With exact readout all capacities of a readout add up to its rank, 45 here [28], so raising the linear memory capacity from 15 to 35 units leaves at most 10 instead of 30 for products of inputs. The fixed-loss control in Table 3 gives a smaller gain than the lifetime-dependent input schedule.

E.4 Measurement (Sec. 3.4)

Weak signals cost measurements: with independent decay at $\gamma = 64$ the previous input moves the state by about 10^{-9}, which gives perfect recall with exact readout and almost none with 10^8 preparations per setting and step. Gentle writing

Table 3: Linear memory capacity $\sum_\tau C_\tau$ of the five-spin reservoir in units (ceiling 45), mean of eight reservoirs, with exact readout or, as recoverable capacity, at a budget of 10^9 reservoir steps per output step (Sec. 3.4).

	independent decay	opposite phases	equal phases
tuned loss, $\eta = 1$	14.5	14.7	31[a]
fixed-loss control[b]	16	25	–
gentle writing	35 ($\gamma = \frac{1}{64}$)	36 ($\gamma = 256$)	–[c]
budget 10^9, $\eta = 1$	13 ($\gamma \approx \frac{1}{8}$)	12 ($\gamma \approx 16$)	13 ($\gamma = 1$)[d]
budget 10^9, gentle	34 ($\gamma = \frac{1}{64}$)[d]	21 ($\gamma \approx 64$)	–[c]

[a] At strong loss. [b] At $\gamma = 1$ and $\eta = \frac{1}{32}$, two reservoirs. [c] The uniform input already writes gently in this channel. [d] Smallest loss rate sampled. The capacity falls as the loss grows.

makes the signal fainter: with the opposite-phase channel it falls from 8×10^{-5} at the full amplitude to 5×10^{-6}, against 7×10^{-5} with independent decay and gentle writing at $\gamma = 1/32$. Restarting each run from $|00000\rangle$, as the replay does, raises these to 4×10^{-5} and 1.2×10^{-4}, about sevenfold for the opposite-phase channel and less than twofold for independent decay. At the full amplitude the signal of the equal-phase channel falls from 2×10^{-4} at $\gamma = 1$ to 4×10^{-7} at $\gamma = 64$, which is why this channel loses capacity as the loss grows once measurements are counted.

The replayed range τ_{90} grows with the memory lifetime, though more slowly: along the gentle line of the opposite-phase channel from about 30 steps at $\gamma = 16$ to 130 at $\gamma = 256$. The budget changes the preferred loss rates within the sampled range: at 10^{10} reservoir steps per output step the best sampled loss rate of the equal-phase channel moves from $\gamma = 1$ to about 16, with 17 units, while the opposite-phase channel with gentle writing peaks near $\gamma = 64$ with 25. Gentle writing becomes worthwhile once each setting is prepared about 2×10^4 times per step with independent decay and 10^5 times with the opposite-phase channel. Below these crossings the full amplitude wins: at 10^4 preparations per setting and step it keeps 11 units with independent decay and 8 with the opposite-phase channel, against 9 and 3 with gentle writing. With the input on one spin, which writes more gently by itself, the crossings move to about 3×10^5 and 4×10^6 preparations. With the full history, eight times more training inputs change the sampled capacity by less than 1.3 units.

For the matched decay-only lifetime of Fig. 4b we find, for every channel and reservoir, the loss rate at which $t_{\mathrm{leak}} = 32$ steps, at weak loss without protected operators and at strong loss with them, and run the reservoir there with $\eta = \sqrt{4/32}$. The write strength is the Euclidean norm of the Pauli coefficients $\mathrm{Tr}(PW)$, over all Pauli strings P, of the change W: the commutator $-i[H_{\mathrm{in}}, \rho_*]$ with $H_{\mathrm{in}} = h \sum_i X_i$ and ρ_* the stationary state, projected spectrally onto the slow modes. These are the protected ones, or without protected operators those decaying at most four times faster than the slowest. The capacity that survives 10^6 preparations rises with the write strength, from 0.3 units for collective decay with equal phases, through one channel or two, to 15 for the opposite-phase channel and 17 for independent decay, and power laws fitted within the channels without and with dark states (dotted in Fig. 4b) have exponents 2.0, as linear accumulation of the write predicts, and 1.5.

E.5 Ordinary noise

Adding independent decay $\sqrt{\gamma_{\mathrm{loc}}}\,\sigma_i^-$ to the equal-phase channel gives $\bar{\kappa} = \kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{loc}}$, where to first order κ_{l} is the trace of the added dissipator on the 99 slow dark modes divided by 99, $140/99 \approx 1.414$. Local dephasing $\sqrt{\gamma_\phi}\,Z_i$ adds $\kappa_\phi\gamma_\phi$ with $\kappa_\phi = 403.2/99 \approx 4.073$, and the full spectra reproduce both coefficients to within 0.2%. A single-input recall time of at least T steps, $\kappa_{\mathrm{c}}/\gamma + \kappa_{\mathrm{l}}\gamma_{\mathrm{loc}} \leq 1/(\Delta t\, T)$, then needs $\gamma/\gamma_{\mathrm{loc}} \geq 4\kappa_{\mathrm{c}}\kappa_{\mathrm{l}}\,\Delta t^2 T^2 \approx 4.2\,T^2$, or $12\,T^2$ for dephasing.

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